
Strategy Evaluation Tool

Assignment Brief

Build a general-purpose tool that ingests a trade-history report, computes a documented set of performance and risk metrics, runs a Monte Carlo robustness analysis, visualises the results, and applies a transparent weighted scoring model that returns a single score out of 100. The tool must be strategy-agnostic and able to save and compare multiple strategy evaluations.

DOCUMENT

OQT-INT-EVAL-01

VERSION

1.0

DURATION

2 weeks

INTERN

START DATE

TECHNICAL REVIEW

Philip Öhlinger, CTO

FIRST CHECKPOINT

End of Day 2

REPORTING

Daily, WhatsApp / email

LANGUAGE

English

1 Purpose and scope

Your task is to build a working **strategy evaluation tool**. It reads a trade-history report from a trading platform, computes performance and risk metrics, runs a Monte Carlo simulation, presents everything visually, and produces a documented score out of 100 that expresses how good the strategy is.

The tool is **strategy-agnostic**. It evaluates whatever trade data it is given and must not assume any specific strategy logic, symbol, or trading style. Someone should be able to feed it any strategy report and get a fair, comparable evaluation.

THE POINT OF THIS ASSIGNMENT

We do not just want a number at the end. We want to see your **evaluation methodology**: which quantities you measure, how you compute them, how you weight them, and why. The reasoning and the mathematics behind each step matter as much as the final score.

2 What you deliver

Pick **one** of the two architectures below and justify your choice in one short paragraph in your Day-2 sketch. You are the engineer here, so decide based on what lets you build a clean, correct engine fastest.

Option A

Python script with a minimal GUI (for example Tkinter, PySimpleGUI, or similar). Everything in one place, quickest to iterate.

Option B

Minimal HTML frontend with a Python engine in the background (for example Flask or FastAPI serving a lightweight page). Cleaner separation, nicer to demo.

Either way:

- Keep the interface **minimalistic**. The engineering weight sits in the evaluation engine, not in UI polish.
- The whole pipeline must be **reproducible**: the same input plus the same random seed produces the same output every time.
- Ship a short technical write-up (a `README` or markdown file in the repo is fine) that documents every formula, every assumption, and every data-cleaning step.

3 Timeline and milestones

Milestone	When	Deliverable
Design sketch + math	End of Day 2	Chosen architecture with a one-paragraph justification, a wireframe of the interface, the full metric list with formulas, the Monte Carlo method written out, the scoring model design (parameters, weights, mapping), and your data-parsing plan. Reviewed by Philip.
Implementation	Day 3 to end of Week 2	Build the tool. Get the Core scope solid and correct first, then move into Stretch as far as you get.
Handover	End of Week 2	Working tool, documentation, and your saved example evaluations. It does <i>not</i> need to be 100% complete.

ON SCOPE

The scope below is deliberately larger than two weeks. That is intentional. We would rather give you too much than too little. Core must be correct and finished. Stretch is where you show range. If Core is done early, keep going.

4 Core vs stretch

CORE must work and be correct by end of Week 2:

- Import and correctly parse the provided platform report.
- Compute the metric set in Section 6.
- Compute drawdown yourself from the trade data and feed it into the score (Section 7).
- Monte Carlo simulation with percentile output and the core charts (Sections 8 and 10).
- Weighted scoring model with at least five parameters, visible sub-scores, and a final score out of 100 (Section 9).
- Save, list, and reload multiple strategy evaluations (Section 11).

STRETCH as far as you get, in roughly this order of value:

- Support a second report format (a MetaTrader MT4/MT5 statement) behind the same interface.
- Reconstruct an approximate equity-based drawdown where the data allows it (Section 7).
- A side-by-side comparison dashboard for multiple saved strategies.
- Rolling metrics, parameter sensitivity, and outlier / robustness tests.
- Export a clean one-page PDF summary per evaluated strategy.

5 Data input · be very specific here

You will receive a `cTrader report.html` based on closed trades. The tool must read reports like this. **Do not assume its structure.** Inspecting the file and verifying every field is part of your Day-1 work, and it is where most of the hidden traps live.

Requirements

- Identify and extract, at minimum: open time, close time, symbol, direction (buy/sell), volume, entry price, exit price, commission, swap, gross profit, net profit, and any position / label / deal ID.
- **Normalise the server timezone.** State which timezone the report uses. This decision affects every time-bucketed metric (daily returns, drawdown timing, monthly breakdown).
- Handle **partial closes**, **multiple simultaneously open positions**, and **hedged positions** correctly. Document how you treat each.
- Separate **net from gross** (with vs without commission and swap) and state explicitly which one each metric uses.

NON-NEGOTIABLE · DATA HONESTY

If a field you need is missing from the report, **say so explicitly** and document the consequence for the affected metric. Never silently substitute, estimate, or invent a value to fill a gap. A tool that reports an honest "not available" is worth far more to us than one that prints a confident wrong number.

6 Evaluation dimensions and metrics

The tool computes a grouped set of metrics. **Every metric must be accompanied by its formula** in your documentation. This is the "mathematical proof of what is evaluated" that we expect: for each quantity, state the estimator, its inputs, and its assumptions. The set below is the minimum.

Group	Metrics	Be precise about
Return	Total return, CAGR, net return after commission and swap.	Whether returns use compounding (equity-scaled) or fixed lot size. This single assumption drives every downstream number, so make it explicit and keep it consistent.
Risk	Maximum drawdown, drawdown duration, average drawdown. See Section 7, this is the centrepiece.	Which equity series the drawdown is measured on, and the closed-trade limitation.
Risk-adjusted	Sharpe, Sortino, Calmar (CAGR / max DD), Profit Factor, Expectancy per trade.	The return series used (per-trade, daily, monthly), the risk-free assumption, and the annualisation factor for Sharpe and Sortino.
Trade stats	Win rate, payoff ratio (avg win / avg loss), longest losing streak, number of trades as sample size.	

Sample size. A great-looking result on 40 trades is not the same as one on 4,000.

Distribution

Histogram of trade returns, skewness, kurtosis, and an outlier test.

The outlier test: remove the top N trades and check whether the strategy still holds up without them.

7 Drawdown · you compute it, it feeds the score

You compute drawdown **directly from the trade data** and derive the corresponding score component from it. You define the methodology. Points you must address:

- Reconstruct the equity or balance curve from the trade sequence, then define maximum drawdown precisely on that curve.
- Document the **limitation** below rather than hiding it behind a clean number.

THE LIMITATION YOU MUST DOCUMENT

The cTrader report contains **closed trades only**. From closed trades you can compute a **balance-based** drawdown, but not the true **equity** drawdown that includes open floating positions. A balance-only drawdown can significantly understate the real drawdown of any strategy that holds losing positions open. State this clearly. Do not present a balance drawdown as if it were the full picture.

This is also the level of mathematical rigour we expect for *every* metric. As a calibration example, maximum drawdown documented properly looks like this:

```
Et = equity (or balance) curve reconstructed from the trade sequence
Pt = running peak = max( Es for all s ≤ t )
Dt = ( Et - Pt ) / Pt    # drawdown at time t, ≤ 0
MaxDD = min( Dt over all t )    # the most negative point
# state whether E is balance-based (closed trades) or equity-based (incl. floating)
```

STRETCH If the report exposes per-position maximum adverse excursion or intra-trade extremes, use them to reconstruct an approximate equity drawdown, and document the approximation and its assumptions.

8 Monte Carlo simulation · the mathematical core

Write the method out **formally**. Your documentation must state what quantity the simulation estimates and under which assumptions. At minimum:

- **Method:** resample the realised trade sequence, by reshuffling order and/or bootstrap with replacement. State clearly which you use and why.
- **Outputs:** the distribution of final equity and of maximum drawdown, percentile bands (for example 5th / 50th / 95th), and a probability-of-ruin figure.
- **Runs and seed:** a defensible number of iterations (for example 5,000 to 10,000) and a **fixed seed** for reproducibility.

THE ASSUMPTION YOU MUST NOT GLOSS OVER

Trades that belong to the same basket, or that overlap in time, are **correlated, not independent**. Naive per-trade shuffling breaks that structure and produces an over-optimistic distribution. Where the data allows, resample at the **basket or position-group level**, and in any case document this assumption and its effect on the result. This is exactly the kind of detail that separates a real evaluation from a pretty chart.

9 Scoring model · score out of 100

- At least **five parameters** feed the score. **Justify every weight**. Do not pick weights arbitrarily, tell us why each one carries the weight it does.
- For each parameter, define how the raw value **maps to a sub-score** (a threshold or a curve). For example, a Profit Factor below 1 maps to 0 and above some ceiling maps to full marks, with a defined shape in between.
- Keep the sub-scores **visible** in a radar chart, not just the final number. We want to see the breakdown.
- Distinguish weighted parameters from **hard knockout criteria**. Some conditions, such as a maximum drawdown beyond a set limit or a Profit Factor below 1, should be able to fail a strategy regardless of the weighted total. Define your knockout rules.

CANDIDATE PARAMETERS (CHOOSE AND JUSTIFY AT LEAST FIVE)

Return quality, risk-adjusted return (Sharpe / Sortino / Calmar), drawdown severity, consistency of returns over time, trade-quality and expectancy, and robustness from the Monte Carlo distribution (for example the 5th percentile outcome rather than the average). You are free to choose your own, as long as you can defend them.

10 Required graphs

CORE

- Equity and balance curve
- Underwater / drawdown curve
- Monte Carlo fan chart with percentile bands
- Histogram of simulated final equity and of simulated max drawdown
- Trade-return distribution
- Radar chart of the scoring sub-scores

STRETCH

- Rolling metrics (rolling Sharpe, rolling win rate)
- Monthly returns heatmap
- Side-by-side comparison charts for multiple strategies

11 Saving and comparing strategies

- Define one consistent **schema** for a saved evaluation: input reference, computed metrics, scoring breakdown, assumptions used, timestamp, and version.
- The tool must **save, list, and reload** multiple evaluations.
- **STRETCH** A comparison view that puts two or more saved strategies side by side.

12 Documentation and reproducibility

- Document every formula, assumption, and cleaning step. A README or markdown file in the repo is fine.
- **Version your assumptions and your scoring weights** so any result can be traced back to the exact configuration that produced it.
- The pipeline must be reproducible: same input plus same seed gives the same output.

13 Non-negotiables · where you must be precise

These are the points where sloppiness quietly invalidates the whole evaluation. Get every one of them right and state it explicitly in your documentation.

Topic	Requirement
Compounding	Stated and consistent across all return metrics.
Drawdown	Methodology defined; the closed-trade limitation documented.
Sharpe / Sortino	Return series, risk-free assumption, and annualisation all stated.
Monte Carlo	Method, number of runs, seed, and the independence / basket-correlation caveat documented.
Scoring	Weights justified, sub-score mapping defined, knockout rules defined.
Data	No invented or silently substituted values; missing fields flagged explicitly.

14 Daily reporting and questions

- Send a **short daily update**, two to four sentences, on what you did that day. WhatsApp or email is fine.
- For questions, use **WhatsApp**, that is the fastest way to get an answer.
- Your Day-2 sketch and your daily updates are reviewed by **Philip Öhlinger (CTO)**.

DAILY UPDATE TEMPLATE (OPTIONAL)

Done today: ... Next: ... Blocked on: ...

Definition of done

- ☐ **Day 2:** sketch and math reviewed and accepted by Philip.
 - ☐ **End of Week 2:** Core scope working and correct.
 - ☐ Stretch scope taken as far as reached.
 - ☐ Documentation present: formulas, assumptions, cleaning steps, scoring weights.
 - ☐ Tool runs reproducibly on the provided report and can save more than one evaluation.
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